AN AMPLE FAMILY OF INVERTIBLE MODULES GIVES THE RESOLUTION PROPERTY

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Definition of an ample family. Morphisms, definition family-ample-invertible-modules (cross-chapter, structural). Finitely many invertible modules with sections whose nonvanishing loci are affine and cover X.
- Finite-cover criterion for the resolution property. lemma-resolution-property-finite-number (carded L26793). Concludes once the complement ideals are quotients of finite locally free modules.

Source reference. Original lemma 0F89: sources/stacks-perfect/pinned-original-source.tex, source lines 9567–9567. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0FXR: focused-seeds/geometry/S9-perf2-resolution-property-ample-family/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a scheme. If X has an ample family of invertible modules (Morphisms, Definition 0FXR), then X has the resolution property.*

Proof. Since X is quasi-compact, there exists n and pairs $(\mathcal{L}_i, s_i)$, $i = 1, \ldots, n$ where $\mathcal{L}_i$ is an invertible $\mathcal{O}_X$-module and $s_i \in \Gamma(X, \mathcal{L}_i)$ is a section such that the set of points $U_i \subset X$ where s_i is nonvanishing is affine and $X = U_1 \cup \ldots \cup U_n$. Let $\mathcal{I}_i \subset \mathcal{O}_X$ be the image of $s_i : \mathcal{L}_i^{\otimes -1} \to \mathcal{O}_X$. Applying Lemma 0F89 we find that X has the resolution property. □

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.